Basketball Action Recognition

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Link to Github: https://github.com/mateotomeho/ActionRecognitionBasketball

1. Project Description - Basketball Action Recognition

This document presents my process in developing a deep learning application for action recognition in basketball. Motivated by the rapid growth of sports analytics and my personal background as a basketball player, the goal is to build a deep learning model that can classify key basketball actions such as shooting, dribbling, passing, and defence. Deep learning is an appropriate tool for this project because basketball action recognition relies on identifying complex spatial and temporal patterns within the video frames. This is something that machine learning methods can not capture as easily as Convolutional Neural Networks, which excel at learning & processing visual and temporal data. Therefore, my model would be able to process video frames to classify basketball actions from players effectively. This will be useful and interesting because this deep neural network could be applied to automate highlight generation, analyze a player's performance and contribute to real-time broadcasting by giving some assistance with the actions occurring during a basketball game.

2. Data Processing

2.1 Data Collection & Processing

For this project, the data was collected from the Space Jam Dataset [1][2], which contains labelled videos of single player basketball actions, including shooting, dribbling, defence and passing. The dataset found is organized into three main components: video files of basketball actions, a label dictionary mapping each action and its corresponding numeral label and an annotations file that associates each video with its respective action label in a list.

To prepare the dataset for training, I performed the following data cleaning and formatting steps: I evaluated the class balance by checking the number of videos per action category and ensuring there were no unlabeled or missing videos (see Appendix - Figure 3). To simplify the training, I organized the dataset into subfolders named for each action : block, pass, run, dribble, shoot, ball in hand, defense, pick, no_action, and walk, and moved each video accordingly, removing the empty "discard" category. Finally, I split each video into 16 frames of 128×128 pixels, creating subfolders for individual videos and their frames, preparing the dataset for both the baseline and primary models.

2.2 Dataset Distribution, Plan for testing & Challenges

The dataset was divided into training (80%), validation (10%) and test (10%) sets to ensure proper evaluation and generalization of the model. I split each video for each category into the 3 different sets while maintaining the class proportions. Also, I compute the class counts for each set to generate class weights to address class imbalance during my model training.

To enhance model generalization and reduce computational complexity, I applied grayscale conversion to all frames. Given that the dataset already contains 593,456 images for all the videos, the grayscale transformation helps reduce input size and memory usage without significantly compromising the information content. Also, it helps the model to focus on motion and shape-based features to recognize the basketball actions. (See Appendix - Figure 4)

For testing, the reserved 10% test set will be used to measure the model's performance on unseen examples. Additionally, I plan to evaluate the model performance on new, unseen data by taking short video clips from NBA player highlights. These new clips would provide a realistic measure of how well the model performs in a new environment. Processing a large number of videos and extracting frames was time-consuming, especially during data organization by labels. To speed up the workflow, I optimized the process by transferring files into updated folders instead of duplicating them.

3. Baseline Model

3.1 Description of the baseline model

As a baseline model, I developed a simple two-dimensional Convolutional Neural Network (CNN) model that can recognize the basketball actions from grayscale images collected from the video clips of a basketball player. In the model, presented in Figure 5, the CNN receives 64x64 grayscale images that are transformed into 2D tensors and normalized before training. (See Appendix - Figure 5)

The model consists of 3 convolutional layers with ReLU activation functions, max-pooling for downsampling and two fully connected layers for classification. Also, I applied a layer of dropout and max pooling after each convolution and for the classification to prevent overfitting. The model is trained using, as a criterion, cross-entropy loss and the Adam optimizer.

Since the dataset contains over 590,000 images, only 10% of the total dataset is used while respecting the proportionality across all classes to maintain balance and reduce the computation time and completely. Also, I took this decision to use the smaller subset due to GPU limitations, as training on the full dataset would be computationally expensive.

3.2 Quantitative and Qualitative Results

From Figure 1, we can see that the simple CNN achieved approximately 51.6% validation accuracy and 50.6% training accuracy, showing that it was able to learn some features between basketball actions but still struggles with complex motion patterns. Therefore, qualitatively, the model can learn basic spatial patterns with basketball movements, but the accuracy is still moderate due to the simplified architecture and limited dataset size.

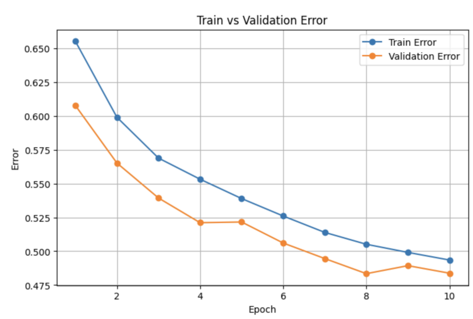

Figure 1: Training and Validation Error of the Baseline Model

3.3 Challenges & Conclusion

Originally, I planned to use an Artificial Neural Network (ANN) as the baseline model, but despite multiple tuning attempts across several architectures, the ANN struggled to learn meaningful visual features from the images and could not exceed around 40% validation accuracy. Therefore, I transitioned to a 2D CNN model to establish a stronger baseline that could better capture spatial patterns and serve as a fair comparison point for more complex models later in the project.

4. Primary Model

4.1 Description of the Primary Model

For my primary model, I implemented a three-dimensional Convolutional Neural Network (3D CNN) to classify the basketball action video clips. The benefit of 3D CNN is that the model convolves over both the spatial (height x width) and temporal (frame sequence dimensions. This process allows to learn spatiotemporal features such as the ball trajectories or player movements across consecutive frames [3]. Each sample consists of a 16-frame grayscale video clip of size 64x64 pixels transferred as a tensor with shape: [Batch = B, Channels = 1, Depth = 16, Height = 64, Width = 64] to classify the 10 basketball actions from the dataset. I kept the same ratio of 80/10/10 for the training, validation and testing sets.

The model architecture consists of three 3D convolutional blocks, increasing the number of channels as follows: 1->16->32->64. Each block includes a 3x3x3 kernel, batch normalization, ReLU activation and 2x2 max pooling. After the 3D convolutions, the resulting feature map is flattened and passed through a fully connected layer to produce a 256-dimensional feature vector. Then, this is mapped to 10 output neurons, one for each class. The training methodology involves using cross-entropy loss as the loss function, Adam optimizer with a learning rate of 1×10^{-3}, batch size of 16 and 20 training epochs.

4.2 Quantitative and Qualitative Results

The model was trained on a 10% subset of the full dataset to demonstrate feasibility. The best validation accuracy achieved was 56.7% at epoch 13, with a corresponding training accuracy of 51-53%, which is above the baseline 2D CNN accuracy of 51.6%, as shown in Figure 2. This demonstrates that this model is effectively learning spatiotemporal patterns, as the training curve shows gradual improvement. However, the validation curve exhibits minor fluctuations, which may indicate slight overfitting in later epochs and should be addressed in future models.

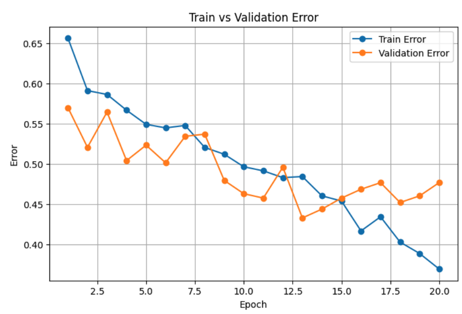

Figure 2: Training and Validation Error of the Primary Model

Regarding qualitative results, the advantage of the 3D CNN is that longer continuous motion, like dribbling, can be classified more accurately than quick actions like blocking because of the temporal context to capture movement patterns. However, confusion can still occur for visually similar actions such as passing and catching, which suggests that more data is needed to improve the model classification.

4.3 Challenges

One main challenge is the limited dataset size. Currently, the model is trained using only 10% of the dataset because of computational constraints, which limits the model's generalization ability. Therefore, increasing the amount of training data seen by the model could improve the accuracy. Also, as observed in Figure 5, slight overfitting occurs in later epochs, to avoid this issue, techniques including early stopping and regularization could control overfitting.

4.4 Conclusion

The 3D CNN successfully demonstrates the feasibility of learning spatiotemporal features from basketball video clips. The primary model achieved a validation accuracy above the baseline, confirming its potential. The primary model still needs further improvements, such as increasing the training data, exploring more advanced 3D CNN architectures and ensuring sufficient training resources to achieve higher accuracy. Also, using pre-trained models could accelerate feature extraction and improve the overall performance.

References

[1] S. Francia, "Classificazione di Azioni Cestistiche mediante Tecniche di Deep Learning", 2018. [Online]. Available: https://www.researchgate.net/publication/330534530_Classificazione_di_Azioni_Cestistiche_mediante_Tecniche_di_Deep_Learning.

[2] S. Francia, "SpaceJam: a Dataset for Basketball Action Recognition," GitHub repository, 2018. [Online]. Available: https://github.com/simonefrancia/SpaceJam. [Accessed: Sep. 26, 2025].

[3] J. Wang, L. Zuo, and C. Cordente Martínez, "Basketball technique action recognition using 3D convolutional neural networks," *Scientific Reports*, vol. 14, no. 1, Art. no. 13156, Jun. 2024. [Online]. Available: https://www.nature.com/articles/s41598-024-63621-8.

Appendix

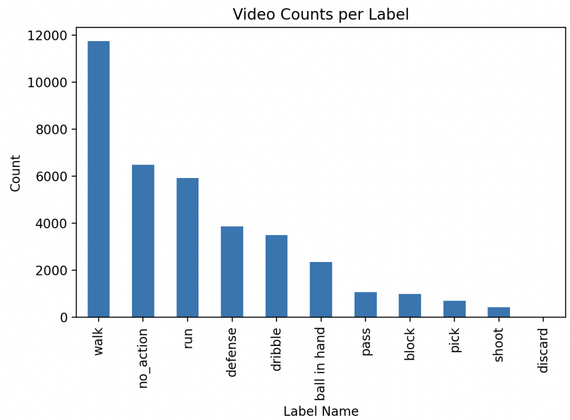

Figure 3: Class distribution and number of videos per category

Dataset Split	Number of Images
Train	474,624
Validation	59,280
Test	59,456
Total	593,360

Figure 4: Dataset Distribution

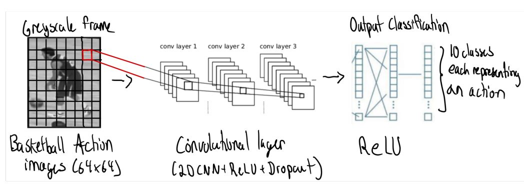

Figure 5: Architecture of the Baseline Model